COMPLETE LEARNING SESSION

Economy Topic 30: Solved Practice Workbook

SOLVED PRACTICE WORKBOOK

LEARNING PATH

CONCEPT
Define precisely

ARGUMENT
Build the case

CRITIQUE
Test the limits

APPLICATION
Write the answer

Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Economy Topic 30: Solved Practice Workbook

Status cutoff: 10 September 2026.

Practice contract: Exactly 32 original MCQs appear before PYQs. Correct answers rotate A -> B -> C -> D eight times. Every option has a question-specific explanation and every MCQ has a unique trap.

BASIC MCQS / REMEDIATION

Q1. Which classification is correct?

A. Animal husbandry covers managed terrestrial livestock and poultry, while capture fisheries harvest a natural stock. B. Aquaculture and capture fishing are identical because both occur in water. C. Milk processing is animal rearing because its raw material is biological. D. All allied-sector activities are secondary-sector activities.

Answer: A

Option explanations:

- **A is correct:** It preserves the biological and economic boundary.
- **B is incorrect:** Aquaculture manages a cultivated stock; capture harvests a wild stock.
- **C is incorrect:** Processing is downstream manufacturing/service activity, not rearing.
- **D is incorrect:** Rearing and fishing are primary activities; downstream functions vary.

UPSC trap: Separate production system from downstream value chain.

Q2. Which statement correctly distinguishes official animal-sector statistics?

A. BAHS is a point-in-time complete enumeration. B. Livestock Census measures stock, while ISS estimates annual product flows. C. ISS measures only export consignments. D. A census count is directly comparable with annual milk tonnage.

Answer: B

Option explanations:

- **A is incorrect:** BAHS is a publication drawing on census, ISS and administrative data.
- **B is correct:** This is the DAHD stock-flow distinction.
- **C is incorrect:** ISS estimates milk, egg, meat and wool output, not exports.
- **D is incorrect:** Stock and flow use different units and reference periods.

UPSC trap: Census, survey and publication are not synonyms.

Q3. Why can livestock smooth rural income?

A. It is immune to drought and disease. B. It always yields a higher return than crops. C. Recurring output and saleable asset value can diversify seasonal crop cash flow. D. It removes all household market risk.

Answer: C

Option explanations:

- **A is incorrect:** Fodder and water shocks can correlate with crop failure.
- **B is incorrect:** Relative return is system- and price-specific.
- **C is correct:** This identifies timing and asset channels without claiming immunity.
- **D is incorrect:** Feed, disease and buyer shocks remain.

UPSC trap: Diversification reduces some covariance; it is not guaranteed insurance.

Q4. Which is the best measure of farm-level livestock profitability?

A. Total national milk output B. Number of animals owned C. Gross sale receipts only D. All revenues plus asset-value change minus paid and imputed costs and losses

Answer: D

Option explanations:

- **A is incorrect:** National output does not reveal household margin.
- **B is incorrect:** Headcount omits productivity and cost.
- **C is incorrect:** Gross receipts omit feed, labour, mortality and finance.
- **D is correct:** This is the complete enterprise-income boundary.

UPSC trap: Gross production is not net producer income.

Q5. At the 10 September 2026 cutoff, which statement is accurate?

A. The 20th Census (2019) remained the latest released population table; 21st Census results were not yet public. B. The 21st Census had published a new national livestock total. C. BAHS 2025 replaced the Livestock Census. D. The Economic Survey is the census authority.

Answer: A

Option explanations:

- **A is correct:** This preserves launch/enumeration and release status.
- **B is incorrect:** No consolidated final 21st-round population table was located.
- **C is incorrect:** BAHS publishes annual products; it does not replace stock enumeration.
- **D is incorrect:** DAHD conducts the census through states.

UPSC trap: A newer round underway is not a released result.

Q6. Which 2024-25 production pairing is correct?

A. Milk 239.30 MT and fish 18.393 MT B. Milk 247.87 MT and fish 19.775 MT C. Eggs 247.87 billion and meat 149.11 MT D. Livestock GVA 19.775% and fish 5.5 MT

Answer: B

Option explanations:

- **A is incorrect:** Those are 2023-24 production values.
- **B is correct:** BAHS 2025 and Economic Survey 2025-26 Table 1.23 support these 2024-25 flows.
- **C is incorrect:** The units and products are reversed and meat is far smaller.
- **D is incorrect:** GVA shares and physical output are being mixed.

UPSC trap: Always attach reference year, product and unit.

Q7. Which statement best captures livestock productivity?

A. Peak daily yield is sufficient. B. Higher herd size always raises household margin. C. Lifetime output, fertility, survival and cost must be read with feed, health and environment. D. Only breed determines realised output.

Answer: C

Option explanations:

- **A is incorrect:** Peak yield can coexist with poor fertility or survival.
- **B is incorrect:** More animals raise maintenance and ecological load.
- **C is correct:** This is the complete lifetime-productivity test.
- **D is incorrect:** Genetics requires complementary inputs.

UPSC trap: Productivity is not headcount or peak yield.

Q8. Which feed-policy statement is most defensible?

A. Fodder has no land opportunity cost. B. Concentrate prices are unrelated to crop markets. C. Breed improvement removes the need for ration planning. D. Fodder seed, storage, residue treatment and commons management must fit local feed balances.

Answer: D

Option explanations:

- **A is incorrect:** Fodder competes for land and water.
- **B is incorrect:** Maize and oilseed markets transmit price shocks.
- **C is incorrect:** Genetic potential fails under poor nutrition.
- **D is correct:** This combines supply, quality, seasonality and resource fit.

UPSC trap: A feed intervention must solve quantity, quality and seasonality.

Q9. What is the correct genetics-by-environment inference?

A. Indigenous and crossbred options must be assessed against feed, heat, disease and lifetime performance. B. Crossbred animals are always economically superior. C. Artificial insemination itself proves higher income. D. Conservation and productivity are mutually exclusive.

Answer: A

Option explanations:

- **A is correct:** This fits breed choice to the production environment.
- **B is incorrect:** Higher potential can carry higher management cost.
- **C is incorrect:** AI is an input stage, not an income outcome.
- **D is incorrect:** Locally adapted genetic resources can support resilient productivity.

UPSC trap: Technology adoption is not realised lifetime performance.

Q10. Why is vaccination a public-good concern?

A. It affects only the vaccinated owner. B. One owner's prevention can lower transmission risk for others. C. It removes the need for surveillance. D. It guarantees eradication after one round.

Answer: B

Option explanations:

- **A is incorrect:** Disease transmission creates external effects.
- **B is correct:** This is the positive network externality.
- **C is incorrect:** Surveillance verifies and guides vaccination.
- **D is incorrect:** Coverage, immunity and eradication are separate.

UPSC trap: Count susceptible population and effective immunity, not doses alone.

Q11. Which climate statement is correct?

A. Methane is a nitrogen compound. B. Lower emissions intensity always lowers total emissions. C. Productivity can reduce emissions per unit while total emissions still rise with expansion. D. Poultry litter creates no nitrogen externality.

Answer: C

Option explanations:

- **A is incorrect:** Methane is CH_4 and contains carbon, not nitrogen.
- **B is incorrect:** Absolute herd/output growth may dominate intensity gains.
- **C is correct:** This preserves intensity versus absolute load.
- **D is incorrect:** Litter can release ammonia and nitrous oxide.

UPSC trap: Intensity and absolute emissions answer different questions.

Q12. Which welfare proposition is strongest?

A. Welfare standards only raise costs. B. Welfare is irrelevant to market access. C. Productivity alone resolves ethical concerns. D. Good housing, handling and transport can improve welfare, survival, quality and compliance, though transition costs need support.

Answer: D

Option explanations:

- **A is incorrect:** Poor welfare also causes stress, injury and loss.
- **B is incorrect:** Standards and buyer requirements affect access.
- **C is incorrect:** Intrinsic welfare obligations remain.
- **D is correct:** This balances benefits with small-producer compliance cost.

UPSC trap: Welfare is both intrinsic and economically consequential.

Q13. Which sequence correctly represents the dairy value chain?

A. Production -> collection/testing -> chilling -> processing -> marketing B. Processing -> breeding -> collection -> feed C. Retail -> milking -> chilling -> testing D. Feed -> export -> veterinary diagnosis -> collection

Answer: A

Option explanations:

- **A is correct:** This follows product and information flow.
- **B is incorrect:** It reverses the production chain.
- **C is incorrect:** Retail cannot precede production.
- **D is incorrect:** Export is not an input-stage substitute.

UPSC trap: Perishability makes sequence and time-temperature control central.

Q14. Which Operation Flood statement is correct?

A. It began in 1991 as a private-dairy programme. B. It began in 1970 and used three phases to build producer cooperatives, services and a milk grid. C. It was limited to breed distribution. D. It eliminated all cooperative-governance risks.

Answer: B

Option explanations:

- **A is incorrect:** NDDB dates launch to 1970.
- **B is correct:** This captures the institutional mechanism and chronology.
- **C is incorrect:** Procurement, processing, marketing and services were central.
- **D is incorrect:** Capture and weak management can persist.

UPSC trap: Operation Flood was institution building, not a cattle subsidy.

Q15. What defines the Anand pattern?

A. One national private processor buying from all states B. A two-tier state bureaucracy C. Village societies, district unions and state federations owned by producers D. An informal trader network without testing

Answer: C

Option explanations:

- **A is incorrect:** Private ownership is a different model.
- **B is incorrect:** The producer institution is not a departmental hierarchy.
- **C is correct:** This is the classic three-tier cooperative structure.
- **D is incorrect:** Informal procurement lacks the stated federation.

UPSC trap: Three tiers describe ownership and functions, not automatic performance.

Q16. Which comparison of dairy models is correct?

A. Private dairies cannot provide extension. B. Producer companies are government departments. C. Informal markets always offer traceability. D. Cooperative, producer-company, private and informal channels differ in ownership, capital, services and bargaining risk.

Answer: D

Option explanations:

- **A is incorrect:** Private firms may bundle services.
- **B is incorrect:** Producer companies are company-form producer institutions.
- **C is incorrect:** Informal channels often have weak formal traceability.
- **D is correct:** This avoids a false binary and identifies economic dimensions.

UPSC trap: Judge governance and competition, not labels alone.

Q17. Which is the accurate current-status description of RGM at the cutoff?

A. It received temporary FY 2026-27 approval through 30 September 2026 or 16th FC approval, whichever earlier, with Rs 800 crore. B. Its only current component is creating new Gokul Grams. C. It is the umbrella for all sheep, goat and pig enterprises. D. Its insemination count equals milk-income impact.

Answer: A

Option explanations:

- **A is correct:** The April 2026 administrative approval supplies this bounded status.
- **B is incorrect:** New Gokul Gram creation is not the current defining design.
- **C is incorrect:** Those non-bovine enterprise components sit mainly under NLM.
- **D is incorrect:** Outputs do not prove conception, productivity or income.

UPSC trap: Date RGM's temporary extension and preserve the genetics boundary.

Q18. Which set correctly states NLM's current three-sub-mission design?

A. Milk procurement, marine fisheries and crop insurance B. Breed development; feed and fodder; innovation and extension C. Gokul Grams, FMD eradication and milk marketing D. Harbours, hatcheries and seafood exports

Answer: B

Option explanations:

- **A is incorrect:** These belong to different schemes and sectors.
- **B is correct:** The January 2025 NLM guidelines retain these three.
- **C is incorrect:** RGM/LHDCP/dairy functions are being merged.
- **D is incorrect:** These are fisheries functions.

UPSC trap: NLM is not a generic label for every animal-sector programme.

Q19. Which component belongs within LHDCP?

A. Milk powder marketing B. Breed multiplication farm finance C. NADCP vaccination plus critical/state disease control and veterinary delivery D. Fishing harbour construction

Answer: C

Option explanations:

- **A is incorrect:** Marketing belongs to dairy value chains.
- **B is incorrect:** Breed multiplication is associated with RGM/NLM and infrastructure instruments.
- **C is correct:** This captures the LHDCP umbrella.
- **D is incorrect:** Harbours belong to fisheries infrastructure.

UPSC trap: Disease-control architecture is distinct from breeding and processing.

Q20. Which statement about AHIDF/IDF is correct?

A. The full fund size is annual cash expenditure. B. Only government departments are eligible. C. It finances crop MSP procurement. D. It supports animal-sector infrastructure; fund size, budget, sanctioned loan, completed asset and utilisation must be separated.

Answer: D

Option explanations:

- **A is incorrect:** The fund is a multi-year financing envelope.
- **B is incorrect:** Eligible entities include private, MSME, FPO, Section 8 and cooperatives.
- **C is incorrect:** MSP procurement is outside its function.
- **D is correct:** This preserves financing-stage discipline.

UPSC trap: Infrastructure approval is not operating capacity or producer benefit.

Q21. Which poultry distinction is correct?

A. Layers produce eggs; broilers are raised mainly for meat. B. Layers and broilers differ only by ownership form. C. Backyard poultry always uses commercial integration. D. Commercial poultry has no biosecurity risk.

Answer: A

Option explanations:

- **A is correct:** This is the biological-output distinction.
- **B is incorrect:** Breed, cycle and output differ.
- **C is incorrect:** Backyard systems can be independent and low-input.
- **D is incorrect:** Density can amplify disease risk.

UPSC trap: Start poultry analysis with output and production system.

Q22. What does poultry integration usually do?

A. It makes the farmer owner of the integrator. B. It coordinates chicks, feed, veterinary inputs and marketing while reallocating contract risks. C. It eliminates mortality and quality disputes. D. It guarantees the spot-market price.

Answer: B

Option explanations:

- **A is incorrect:** Ownership does not automatically transfer.
- **B is correct:** This states both coordination and risk allocation.
- **C is incorrect:** Mortality attribution is a common dispute.
- **D is incorrect:** Payment may be a growing charge, not spot price.

UPSC trap: Integration reallocates risk; it does not abolish it.

Q23. Why are sheep and goat systems important in drylands?

A. They require no health services. B. They cannot use commons. C. They can use dispersed browse and mobility while providing multiple outputs and liquid-asset value. D. They are always environmentally benign at any stocking rate.

Answer: C

Option explanations:

- **A is incorrect:** Disease control remains necessary.
- **B is incorrect:** Commons and routes often underpin production.
- **C is correct:** This captures their livelihood logic.
- **D is incorrect:** Overstocking can degrade grazing resources.

UPSC trap: Pastoral mobility is an adaptive production strategy.

Q24. Which piggery statement is accurate?

A. Disease does not create correlated losses. B. Markets are identical across India. C. Waste management is irrelevant. D. Fast reproduction can support enterprise growth but raises the importance of feed, biosecurity, reporting and regional demand.

Answer: D

Option explanations:

- **A is incorrect:** Outbreaks can destroy many herds.
- **B is incorrect:** Cultural and regional demand differs.
- **C is incorrect:** Waste and sanitation affect disease and externalities.
- **D is correct:** This balances potential with constraints.

UPSC trap: Do not recommend piggery without market and disease context.

Q25. Which distinction is correct?

A. Capture fisheries harvest wild stocks; aquaculture farms managed stocks. B. Aquaculture is always marine. C. Capture fisheries have no common-pool problem. D. All fish production is measured by Livestock Census.

Answer: A

Option explanations:

- **A is correct:** This is the core production-system distinction.
- **B is incorrect:** Aquaculture can be inland, marine or brackish.
- **C is incorrect:** Open access can cause a race to fish.
- **D is incorrect:** Fish output comes from fisheries statistics, not livestock stock enumeration.

UPSC trap: Total fish output can combine very different systems.

Q26. Which federal statement is correct?

A. States regulate the entire EEZ exclusively. B. States regulate fisheries within territorial waters, while Union competence covers fishing beyond territorial waters. C. CAA regulates all inland livestock farms. D. Territorial waters extend to 200 nautical miles.

Answer: B

Option explanations:

- **A is incorrect:** Beyond-territorial-water fishing is a Union competence.
- **B is correct:** This preserves the constitutional and maritime boundary.
- **C is incorrect:** CAA regulates coastal aquaculture.
- **D is incorrect:** Territorial waters extend 12 nautical miles; the EEZ extends farther.

UPSC trap: Do not merge territorial waters and the EEZ.

Q27. Why can MEY occur at lower effort than MSY?

A. MEY ignores harvesting cost. B. MSY maximises profit by definition. C. As stock falls, harvesting cost rises, so profit may peak before biological catch does. D. MEY is an equity rule.

Answer: C

Option explanations:

- **A is incorrect:** MEY explicitly includes revenue and cost.
- **B is incorrect:** MSY is a biological yield benchmark.
- **C is correct:** This is the standard bioeconomic mechanism.
- **D is incorrect:** Distribution is a separate policy objective.

UPSC trap: MSY, MEY, equity and ecosystem safety are distinct.

Q28. What is the correct RAS treatment statement?

A. Biofilters mechanically remove all solids. B. Biofilters eliminate every pathogen and nitrate. C. Biofilters increase phosphorus for fish. D. Mechanical filtration removes solids, while microbes in biofilters nitrify ammonia within a wider treatment system.

Answer: D

Option explanations:

- **A is incorrect:** Solids removal is principally mechanical.
- **B is incorrect:** Disinfection and nitrate management remain.
- **C is incorrect:** Excess phosphorus is a pollution concern.
- **D is correct:** This assigns the correct function to each stage.

UPSC trap: A biofilter is not a complete water purifier.

Q29. Which PMMSY status statement is correct at the cutoff?

A. Original approval covered 2020-21 to 2024-25; FY 2026-27 BE provides Rs 2,500 crore, but that budget line is not a newly verified long-term tenure. B. The scheme ended with no later budget provision. C. It began before Blue Revolution. D. Its allocation proves export and income targets were achieved.

Answer: A

Option explanations:

- **A is correct:** This reconciles original approval and current budget evidence.
- **B is incorrect:** Demand No. 43 carries FY 2026-27 provision.
- **C is incorrect:** Blue Revolution preceded PMMSY.
- **D is incorrect:** Allocation is not outcome evidence.

UPSC trap: Separate original tenure, current budget and measured outcomes.

Q30. Which statement describes FIDF accurately?

A. It is a grant-only crop scheme. B. It used a Rs 7,522.48 crore fund and was extended through 2025-26 for concessional fisheries infrastructure finance. C. It permanently guarantees every fisheries loan. D. Its approved fund size equals completed harbour value.

Answer: B

Option explanations:

- **A is incorrect:** FIDF is infrastructure finance, not a crop grant.
- **B is correct:** This matches the Department of Fisheries annual report.
- **C is incorrect:** Credit guarantee and eligibility are conditional.
- **D is incorrect:** Fund, sanction, disbursement and completed asset differ.

UPSC trap: No post-2025-26 extension should be invented.

Q31. Which KCC statement is correct?

A. KCC covers only crop inputs. B. The entire Rs 5 lakh limit automatically receives interest subvention. C. KCC was extended to animal husbandry and fisheries for working capital; limit and interest-benefit ceiling must be separated. D. KCC is a livestock insurance policy.

Answer: C

Option explanations:

- **A is incorrect:** The 2018 extension covers allied sectors.
- **B is incorrect:** The Fisheries Annual Report distinguishes the broader limit from the fisheries IS/PRI ceiling.

- **C is correct:** This is the correct dated design principle.
- **D is incorrect:** Credit and insurance are separate instruments.

UPSC trap: Card limit, subsidised-interest amount and actual borrowing are not the same.

Q32. Which policy package best fits animal-rearing economics?

A. Maximise animal numbers and relax all standards. B. Fund only processing plants. C. Rely only on superior germplasm. D. Combine feed, preventive health, suitable genetics, finance/insurance, infrastructure, competitive markets and sustainability safeguards.

Answer: D

Option explanations:

- **A is incorrect:** Headcount can worsen cost and externalities.
- **B is incorrect:** Plants fail without supply, utilisation and markets.
- **C is incorrect:** Genetics is one complement.
- **D is correct:** This is the complete complementary policy framework.

UPSC trap: A scheme list is not a systems policy.

PYQS AND ANSWER PRACTICE

Verified PYQ 1 - GS-III 2015 - 10 marks, 150 words

> Livestock rearing has a big potential for providing non-farm employment and income in rural areas. Discuss suggesting suitable measures to promote this sector in India.

Demand: Establish employment and income channels, identify binding constraints and suggest matched measures.

Model answer:

Livestock rearing is an allied primary activity, but it generates extensive non-farm work in feed supply, veterinary care, collection, transport, processing and retail. Milk, eggs and short-cycle poultry provide cash between crop harvests; goats and cattle also serve as assets. It can include land-poor households and women, use crop residues, improve diets and diversify weather and crop-price risk.

Potential is constrained by fodder scarcity, low lifetime productivity, weak veterinary reach, disease, costly credit, poor insurance, fragmented surplus, buyer power and inadequate chilling or processing. Women may supply labour without controlling animals, cooperative membership or payments.

Policy should sequence local fodder plans, preventive health and suitable genetics; align KCC and insurance with biological cycles; strengthen cooperatives/FPOs and fair contracts; invest in chilling, testing, processing and traceability; and enforce One Health, welfare and manure safeguards. The objective should be value and net income per healthy animal, not herd expansion.

Why this earns marks: It distinguishes rearing from downstream jobs, balances potential and constraints, and links each reform to a diagnosed mechanism.

Verified PYQ 2 - GS-III 2019 - 10 marks, 150 words

> How far is Integrated Farming System helpful in sustaining agricultural production?

Demand: Explain the mechanism, assess the extent of help and qualify the verdict.

Model answer:

Integrated Farming System deliberately combines crops with livestock, poultry, fisheries or horticulture so that material, labour and income flows complement one another. Crop residues become feed; manure or biogas slurry returns nutrients to soil; pond water or silt can support crops; and multiple products spread labour and cash flow across seasons.

IFS can sustain production by reducing purchased-input dependence, improving soil organic matter, recycling water and nutrients, diversifying climatic and price risk, and supplying household milk, eggs or fish. It is especially useful where small farms can intensify output per unit land without monocrop dependence.

However, mere co-location is not integration. Excess manure or pond nutrients can become pollution; disease can move across enterprises; capital, labour, water, skill and market needs may exceed household capacity. Thus IFS is substantially helpful when location-specific design, veterinary and extension support, credit, sanitation and collective marketing keep recycling within ecological absorptive limits.

Why this earns marks: It answers 'how far' with mechanism, benefits, constraints and a conditional verdict.

Verified PYQ 3 - GS-III 2022 - 15 marks, 250 words

> What is Integrated Farming System? How is it helpful to small and marginal farmers in India?

Demand: Define IFS, centre the smallholder mechanism and include implementation limits.

Model answer:

Integrated Farming System is the planned combination of crop and allied enterprises so that the output or waste of one becomes an input for another and the household receives diversified food, work and cash flow.

For small and marginal farmers, it improves resource productivity: crop residues feed cattle or goats; manure and biogas slurry support soil fertility; pond water and silt can complement crops; and poultry, eggs, milk, fish or vegetables add frequent returns to seasonal crop income. Enterprise diversity spreads production and price risk, uses family labour through the year, reduces some purchased-input costs and improves access to protein and micronutrients. Livestock also provides a saleable asset during emergencies. Producer groups can aggregate milk, fish or eggs and share chilling, feed and veterinary services.

The benefits are conditional. Small plots, insecure tenancy, fodder and water scarcity, high start-up cost, women's unpaid workload, disease transmission, weak veterinary support and thin markets can make a complex system unviable. Nutrient recycling becomes pollution when manure or aquaculture loads exceed absorptive capacity.

Therefore, extension should design agro-climate-specific combinations, sequence feed and health support, align KCC and insurance with biological cycles, secure commons and water access, strengthen women's asset and payment control, and connect FPO/cooperative aggregation to cold chains and markets. IFS sustains smallholders when integration is functional, not when enterprises are merely placed together.

Why this earns marks: The answer defines, explains smallholder channels, recognises gender and ecological constraints, and supplies implementable measures.

Objective PYQ official-key discipline

- **2019 Prelims - nitrogen compounds from agriculture/livestock:** concept route covers ammonia and nitrous oxide from manure/litter; methane is not a nitrogen compound. **Answer withheld pending official UPSC key.**
- **2023 Prelims - biofilters in recirculating aquaculture:** concept route separates microbial nitrification from mechanical solids removal and complete purification. **Answer withheld pending official UPSC key.**
- **2024 Prelims - primary activities:** dairy farming is primary production; processing is secondary. The audited route did not carry a matched question-option-key bundle into this package, so no letter is inferred.
- **2025 Prelims - Rashtriya Gokul Mission and indigenous cattle:** the official Set-A key exists locally, but this owner does not contain an authenticated question-option mapping. No answer letter is inferred.
- **2026 Prelims - FAO Blue Transformation:** the locally held Set-A key is provisional. **Answer withheld because the locally held key is provisional.**

ORIGINAL MAINS 1 - 10 MARKS - 150 words

Question: Explain how animal husbandry diversifies income and risk for small rural households.

Model answer:

Animal husbandry diversifies rural livelihoods through timing, assets and joint products. Milk and eggs generate frequent cash between crop harvests; goats, sheep or cattle provide saleable asset value; manure, offspring and household food add returns not captured by one product price. Crop residues can become feed, linking livestock to mixed farming, while veterinary, collection and processing create non-farm jobs.

The effect is strongest for smallholders and land-poor households when entry scale is modest and women control animals, membership and payments. Yet it is not automatic insurance. Drought raises fodder and water cost; disease creates correlated losses; feed inflation and a single buyer can compress margins; distress sale sacrifices future output.

Hence diversification policy should combine feed reserves, preventive health, suitable breeds, KCC, responsive insurance, producer aggregation and transparent markets. Success should be measured by stable net household income and control over proceeds, not by enterprise count.

ORIGINAL MAINS 2 - 10 MARKS - 150 words

Question: Compare cooperative, producer-company and private dairy models from the producer's perspective.

Model answer:

Dairy organisation determines who owns procurement, bears risk and receives value. The Anand cooperative model federates village societies, district unions and state marketing bodies; it can return residual value and bundle feed, breeding and veterinary services. Its weaknesses are political capture, weak professional management and delayed payment.

A producer company retains producer shareholding within a company form, potentially combining collective ownership with managerial flexibility. It still needs capital, credible boards and member accountability. A private dairy can mobilise investment, technology, brands and procurement competition, but investor control may leave small suppliers with weak bargaining power or selective collection.

No form is automatically superior. Producers benefit where testing is transparent, payments are timely, services are reliable, multiple buyers contest the milk shed and members can audit or exit. Regulation should protect contracts and quality without suppressing investment. The test is producer share and control, not legal label.

ORIGINAL MAINS 3 - 15 MARKS - 250 words

Question: Why should livestock policy prioritise productivity over herd expansion? Discuss the climate trade-off.

Model answer:

Herd expansion raises total output only if additional animals receive adequate feed, water, health care and market access. Where these are scarce, more animals can lower per-animal nutrition, fertility and lifetime yield while increasing household maintenance cost, disease exposure and pressure on commons.

Productivity policy instead targets value per healthy animal through suitable genetics, balanced ration, preventive vaccination, reproductive management, heat-resilient housing and lower mortality. Higher lifetime output spreads maintenance emissions and costs over more milk or meat and may reduce methane intensity per unit. Better fertility can also reduce the number of non-productive animals needed for a given output.

The climate gain is conditional. Total methane can still rise if herd or aggregate output expands faster than intensity falls. Concentrate feed may shift land, water and emissions elsewhere; manure-to-biogas systems can leak; high-yield breeds can be heat- and disease-sensitive. Welfare and indigenous genetic diversity must not be sacrificed for peak yield.

Policy should therefore track per-animal and lifetime productivity, net margin, absolute herd and emissions, feed footprint, manure management and heat resilience together. RGM genetics must be paired with NLM fodder and LHDCP health; producer prices and processing must absorb quality output. The objective is fewer unproductive losses and higher sustainable value, not a headcount race.

ORIGINAL MAINS 4 - 15 MARKS - 250 words

Question: Evaluate the contract-integration model in Indian poultry.

Model answer:

Poultry integration emerges because chick quality, feed formulation, veterinary protocol, batch timing and marketing are tightly interdependent. The integrator commonly supplies chicks, feed, medicine, technical advice and market access; the farmer supplies the shed, utilities and labour and receives a growing charge.

The model can lower input-search costs, standardise biosecurity, spread technical knowledge and shield growers from part of the spot output-price risk. Coordinated placement and processing also reduce market mismatch in a short-cycle industry. These gains are valuable where independent farmers lack working capital or assured buyers.

However, risk is reallocated, not removed. Farmers may borrow for asset-specific sheds, while contracts permit quality deductions or assign mortality and utility risks asymmetrically. Dependence on one integrator weakens exit and bargaining. Feed and disease shocks may change placement volumes, and performance standards can be opaque. Concentrated production also raises regional biosecurity and waste externalities.

Fair integration requires plain-language written contracts, independent weighing and quality records, disclosed mortality formulas, time-bound payment, accessible dispute resolution, minimum biosecurity and welfare standards, and alternative buyers or producer organisations. Credit assessment should test contract durability rather than only shed collateral. Integration is efficient when coordination gains are shared; it becomes exploitative when the integrator controls information, standards and exit without accountability.

ORIGINAL MAINS 5 - 20 MARKS - 250 words

Question: Analyse the economic and ecological challenges of expanding India's fisheries sector.

Model answer:

India's fisheries economy combines inland and marine systems, capture of wild common-pool stocks and managed aquaculture. Economic Survey 2025-26 records 19.775 million tonnes of fish production in 2024-25, but aggregate growth can conceal local stock depletion, disease or weak first-sale margins.

Capture fisheries face a race-to-fish problem: individual effort reduces the stock available to others, encouraging overcapacity, juvenile catch and profit dissipation. MSY is only a biological benchmark; MEY, ecosystem safety and equitable access may require lower effort. Seasonal closures and gear rules therefore need science, enforcement and livelihood support. Governance is federal: states regulate within territorial waters, while Union competence covers fishing beyond them.

Aquaculture raises output through seed, feed, water control and density, but intensification increases disease correlation, energy use, effluent, salinity and tenure conflict. RAS biofilters manage ammonia within a wider treatment chain; they are not complete purification. Both systems need ice, cold chain, hygienic landing, processing, traceability and SPS compliance.

PMMSY's original 2020-21-2024-25 approval and FY 2026-27 budget provision must be distinguished; FIDF's approved extension ended in 2025-26. Reform should strengthen stock assessment and co-management, certified seed and aquatic health, climate-resilient infrastructure, FFPOs, KCC and insurance, transparent auctions, habitat protection and export diversification. The objective is durable value per unit of ecosystem, not output alone.

ORIGINAL MAINS 6 - 20 MARKS - 250 words

Question: Design an integrated policy framework for India's livestock, dairy, poultry and fisheries economy.

Model answer:

India's allied-sector policy should begin with the household enterprise margin and the biological resource, not a headcount target. DAHD reports livestock at 30.87% of agriculture GVA in 2023-24; 2024-25 production reached 247.87 million tonnes of milk, 149.11 billion eggs and 10.50 million tonnes of meat, while fish output was 19.775 million tonnes. These figures establish scale, not producer welfare.

First, secure feed, fodder, water and suitable genetics; RGM must pass a genetics-by-environment test and NLM must close fodder and non-bovine enterprise gaps. Second, treat vaccination, surveillance, diagnostics, compensation and

One Health as public goods through LHDCP and strong state veterinary capacity. Third, align KCC, term finance and insurance with biological cycles. Fourth, build producer-controlled or competitive aggregation, transparent testing, chilling, landing, processing and traceability through dairy institutions, IDF/AHIDF, PMMSY and appropriate infrastructure finance.

Fifth, regulate market power and contracts: cooperatives, producer companies, private dairies, poultry integrators and exporters must disclose quality, rejection and payment rules. Sixth, internalise methane, manure, AMR, welfare, aquaculture effluent and capture-fishery stock costs. Women's ownership, payment control, pastoral access and small-fisher rights should be outcome indicators.

Finally, evaluate net margin, lifetime productivity, survival, producer share, disease incidence and ecological load rather than allocations or assets. Topic 14 supplies input-credit depth and Topic 15 processing economics. A complete framework joins feed, health, genetics, infrastructure, markets and sustainability.